Experimental Lattice Attacks Analysis: Module-LWE and Rank-Dimension Tradeoffs

Navid Abapour
Surrey Centre for Cyber Security
University of Surrey, UK
n.abapour@surrey.ac.uk

Hamidreza Amini Khorasgani
Department of Computer Science
Purdue University, USA
haminikh@purdue.edu

***Abstract*—Module Learning with Errors (MLWE) underlies the lattice-based standards selected by NIST for post-quantum cryptography, specifically the ML-KEM (Kyber) and ML-DSA (Dilithium) schemes for module-lattice-based key encapsulation and digital signatures. In these cases, instead of specifying only the ring dimension n, we consider both the ring dimension n and the module rank k; the total dimension is then given by $d = n{\cdot}k$. Even with d fixed and ignoring subfield attacks, the impact of the choice of the (n, k) decomposition on security has remained unexplored from a concrete security perspective. Theoretical reductions provide worst-case hardness guarantees for MLWE. However, there are no concrete experiments across different rank–dimension decompositions to evaluate actual security or to provide insight into the trade-offs that arise when choosing a particular module architecture based on implementation considerations in practice.**

In this paper, we report on the security of MLWE for 32 different settings across six different dimensions ranging from $d = 256$ to $d = 1536$. Our results demonstrate that for a given value of d, all possible pairs of (n, k) yield identical estimates for generic attacks on MLWE. The costs for both primal attacks on the unique Shortest Vector Problem and Bounded Distance Decoding, as well as dual attacks on MLWE, depend only on d. Our results on small secret distribution show how the Centered Binomial Distribution sampling incurs a small cost of less than 0.3 bits for MLWE attacks, while sampling from a binary distribution leads to a security loss of between 15 to 41 bits depending on the dimensions. In our examination of overstretched MLWE, we observe that, for an error distribution with standard deviation σ and ring modulus q, security is compromised when $\log_2(q/\sigma^2) \approx 15$. These results not only support the flexibility of MLWE for use in standardized schemes, but also point to specific bands that could be avoided in future schemes where these techniques can be applied.

I. Introduction

The Learning With Errors (LWE) problem, introduced by Regev [1], has become the dominant foundation for lattice-based cryptography. Its variants, Ring-LWE [2] and MLWE [3], enable efficient constructions while preserving worst-case hardness guarantees. The standardization of ML-KEM [4], [5] and ML-DSA [6] by NIST, both based on MLWE, has been a transition toward a post-quantum secure communications infrastructure.

MLWE has a middle ground in the LWE hierarchy [7]; for a cyclotomic ring $\mathcal{R} = \mathbb{Z}[X]/(X^n + 1)$ and module rank k, the problem operates over $\mathcal{R}_q^k$, where $\mathcal{R}_q = \mathbb{Z}_q[X]/(X^n + 1)$, with total dimension $d = n \cdot k$. When $k = 1$, MLWE reduces to Ring-LWE; when $n = 1$, it reduces to standard LWE. This parameterization admits multiple decompositions of any target dimension: for instance, $d = 768$ can be realized as $(n, k) \in \{(768, 1), (384, 2), (256, 3), (192, 4), (128, 6), (96, 8)\}$. The ML-KEM standard adopts $n = 256$ with $k \in \{2, 3, 4\}$, a choice driven primarily by implementation considerations, the fixed ring dimension enables a single optimized Number Theoretic Transform (NTT) implementation across all security levels. Here, the natural question arises: Does this choice of (n, k) decomposition affect security? Does the algebraic structure (rank vs. ring dimension) affect hardness against lattice attacks, or does only the product $n{\cdot}k$ matter practically? The theoretical reductions of Langlois and Stehlé [3] establish that MLWE is at least as hard as Ring-LWE for the same ring dimension n, but they do not address the concrete security of different decompositions at fixed total dimension. This gap between asymptotic theory and concrete security estimates motivates our investigation in this work.

For the domain of concrete security for LWE-based schemes, Albrecht et al. [8] introduced the lattice-estimator framework, which has become the standard tool for parameter selection and is used in Zama's Concrete Compiler [9], [10]. The report by MATZOV [11], [12], [13] improved the analysis for dual attack, which provides new estimates for ML-KEM. Bai and Galbraith [14] showed that small secrets are more efficient for lattice embedding, while Albrecht [15] demonstrated that the dual attack can exploit small secrets more efficiently than previously understood.

In the context of LWE-based systems, Applebaum et al. [16] showed that if the entries of the secret are drawn from the same error distribution, then there is no loss of security with this approach. The security of binary and ternary secrets has been well-studied: while earlier works indicated a loss of security with small secrets [17], a recent work by Pouly and Shen [18] indicates that binary LWE is actually polynomially equivalent to standard LWE with appropriate scaling of parameters. However, the impact of small secrets on the concrete security of the system remains unclear and requires empirical analysis.

For NTRU, the dense sublattice attack of Kirchner and

Fouque [19], [20] demonstrated that overstretched parameters admit polynomial-time key recovery, fundamentally constraining the parameter space. No analogous attack is known for MLWE, but the overstretched regime where q/σ^2 is large still exhibits degraded security against standard lattice reduction.

In this work, first, we analyze the rank-dimension tradeoff of MLWE by considering 32 different (n, k) pairs across six total dimensions $d \in \{256, 512, 768, 1024, 1280, 1536\}$. We show that the three generic lattice attacks we consider, primal unique-SVP (uSVP), primal Bounded Distance Decoding (BDD), and dual hybrid [21], [22], yield approximately the same cost for every decomposition of a given d. For $d = 512$, for instance, configurations ranging from $(512, 1)$ to $(32, 16)$ all achieve 143.7 bits of primal and 139.6 bits of dual security, with the same BKZ block size β and the same optimal attack-lattice dimension. Second, we measure the security loss incurred by eight different secret distributions across four dimensions. Using the Centered Binomial Distribution CBD_η (the sampler adopted in ML-KEM) with parameter $\eta = 3$ costs less than 0.3 bits, while CBD_1 costs 9.5 to 25.4 bits; ternary secrets over $\{-1, 0, 1\}$ cost 7.0 to 18.6 bits and binary secrets over $\{0, 1\}$ cost 15.3 to 41.4 bits; sparse ternary secrets with Hamming weight $h = 64$ incur the largest loss, 62.7 bits at $d = 1280$. Third, we identify the overstretched regime, in which security decreases linearly with $\log_2(q/\sigma^2)$ on the interval $[10, 20]$ at a rate of 6 to 8 bits per unit, and we locate the threshold $\log_2(q/\sigma^2) \approx 15$ above which dual attacks become dominant. ML-KEM uses $q = 3329$ and $\sigma \approx 1.22$, giving $\log_2(q/\sigma^2) \approx 11.5$, safely below this threshold. Fourth, we compare MLWE with NTRU at equivalent dimensions: ML-KEM-512 and NTRU-509 reach comparable security at NIST Level 1 (136.0 vs. 137.3 bits), but MLWE offers greater parameter flexibility and a more developed analytical tool-set.

II. Preliminaries

We denote by $\mathbb{Z}_q = \mathbb{Z}/q\mathbb{Z}$ the integers modulo q. Vectors are bold lowercase $(\mathbf{a}, \mathbf{s})$, matrices bold uppercase $(\mathbf{A}, \mathbf{B})$, and $\|\mathbf{x}\|$ denotes Euclidean norm. The notation $x \leftarrow\$ \mathcal{D}$ indicates sampling from distribution $\mathcal{D}$, and $x \leftarrow\$ S$ denotes uniform sampling from set S. Logarithms are base 2.

A *lattice* $\Lambda \subset \mathbb{R}^n$ is a discrete additive subgroup generated by linearly independent vectors $\mathbf{b}_1, \ldots, \mathbf{b}_d \in \mathbb{R}^n$: $\Lambda = \mathcal{L}(\mathbf{B}) = \left\{\sum_{i=1}^{d} z_i \mathbf{b}_i : z_i \in \mathbb{Z}\right\}$, where $\mathbf{B} = [\mathbf{b}_1 | \cdots | \mathbf{b}_d]$ is the basis matrix of rank d, with determinant $\det(\Lambda) = |\det(\mathbf{B})|$ for full-rank lattices. The first successive minimum $\lambda_1(\Lambda)$ is the norm of the shortest nonzero vector; finding it constitutes the Shortest Vector Problem (SVP).

A. Learning With Errors

The LWE problem [1] is parameterized by dimension n, modulus q, and error distribution χ over $\mathbb{Z}$.

Definition 1 (LWE Distribution)**.** For secret $\mathbf{s} \in \mathbb{Z}_q^n$, sample $\mathbf{a} \leftarrow\$ \mathbb{Z}_q^n$, $e \leftarrow\$ \chi$, and output $(\mathbf{a}, \langle \mathbf{a}, \mathbf{s} \rangle + e \mod q)$.

Definition 2 (Search-LWE)**.** Given m samples from $A_{\mathbf{s},\chi}$ for uniform $\mathbf{s} \leftarrow\$ \mathbb{Z}_q^n$, find $\mathbf{s}$.

Definition 3 (Decision-LWE)**.** Distinguish m samples from $A_{\mathbf{s},\chi}$ from uniform samples over $\mathbb{Z}_q^n \times \mathbb{Z}_q$.

LWE hardness follows from worst-case reductions from GapSVP and SIVP [1], [23].

Let $\mathcal{R} = \mathbb{Z}[X]/(X^n + 1)$ for n a power of 2, and $\mathcal{R}_q = \mathcal{R}/q\mathcal{R} \cong \mathbb{Z}_q[X]/(X^n + 1)$.

Definition 4 (Ring-LWE [2])**.** For secret $s \in \mathcal{R}_q$ and error distribution χ, sample $a \leftarrow\$ \mathcal{R}_q$, $e \leftarrow\$ \chi$, output $(a, a \cdot s + e) \in \mathcal{R}_q^2$.

Definition 5 (MLWE [3])**.** For parameters (n, k, q) and secret $\mathbf{s} \in \mathcal{R}_q^k$, sample $\mathbf{a} \leftarrow\$ \mathcal{R}_q^k$, $e \leftarrow\$ \chi$, output $(\mathbf{a}, \langle \mathbf{a}, \mathbf{s} \rangle + e) \in \mathcal{R}_q^k \times \mathcal{R}_q$.

The total dimension is $d = n \cdot k$; Ring-LWE is $k = 1$, standard LWE is $n = 1$. Worst-case hardness guarantees for MLWE are established in [3].

B. Error Distributions

The discrete Gaussian $D_{\mathbb{Z},\sigma}$ assigns probability $\propto \exp(-x^2/2\sigma^2)$ to each $x \in \mathbb{Z}$. The centered binomial CBD_η samples $a_i, b_i \leftarrow\$ \{0, 1\}$ and outputs $\sum_{i=1}^{\eta}(a_i - b_i) \in \{-\eta, \ldots, \eta\}$ with variance $\eta/2$; ML-KEM [4] uses this for constant-time sampling. Ternary and binary distributions sample from $\{-1, 0, 1\}$ and $\{0, 1\}$ respectively; sparse ternary fixes Hamming weight h.

C. Security Estimation

We use the lattice-estimator [8], adopted for ML-KEM [4], [24] and ML-DSA [6] parameter selection, with BKZ reduction [25], [26] costing $2^{0.292 \cdot \beta + o(\beta)}$ via sieving [27], [28] technique. The primal attack [29] embeds LWE samples $(\mathbf{A}, \mathbf{b} = \mathbf{A}^T\mathbf{s} + \mathbf{e})$ into a lattice $\Lambda = \left\{(\mathbf{x}, \mathbf{y}, z) \in \mathbb{Z}^m \times \mathbb{Z}^n \times \mathbb{Z} : \mathbf{x} = \mathbf{A}^T\mathbf{y} + z\mathbf{b} \mod q\right\}$ containing the short vector $(\mathbf{e}, -\mathbf{s}, 1)$; BKZ recovers it for sufficient β. The dual hybrid attack [15], [30], [11] finds short $\mathbf{v}$ in the dual lattice [31] such that $\langle \mathbf{v}, \mathbf{b} \rangle$ reveals error information, combining lattice reduction with combinatorial search and FFT-based distinguishing.

We report security as $\log_2(\mathrm{Cost})$ bits using the MATZOV model [11], [12], [13]: $\mathrm{Cost} = 2^{0.292 \cdot \beta + o(\beta)} \cdot \mathrm{poly}(d)$.

III. Module Structure Tradeoffs

For a fixed total dimension $d = n \cdot k$, multiple (n, k) decompositions yield MLWE instances with identical dimension but differing structural properties. We examine configurations across six target dimensions: $d = 256$ (lightweight), $d \in \{512, 768, 1024\}$ (NIST Levels 1, 3, 5 corresponding to ML-KEM), and $d \in \{1280, 1536\}$ (extended security). For each d, we consider all valid factorizations where n is a power of 2 (for number theoretic transform (NTT) compatibility) and $k \geq 1$. Table I summarizes the 32 configurations analyzed.

TABLE I: MLWE configurations under analysis.

d	**Configurations** (n, k)
256	$(256, 1), (128, 2), (64, 4), (32, 8)$
512	$(512, 1), (256, 2), (128, 4), (64, 8), (32, 16)$
768	$(768, 1), (384, 2), (256, 3), (192, 4), (128, 6), (96, 8)$
1024	$(1024, 1), (512, 2), (256, 4), (128, 8), (64, 16)$
1280	$(1280, 1), (640, 2), (256, 5), (128, 10)$
1536	$(1536, 1), (512, 3), (384, 4), (256, 6)$

When considering lattice attacks against MLWE, the algebraic structure is unrolled into standard LWE over $\mathbb{Z}_q$: a sample $(\mathbf{a}, b) \in \mathcal{R}_q^k \times \mathcal{R}_q$ corresponds to coefficient vectors of dimension $d = n \cdot k$, which yields attack lattices of dimension approximately $2d$ [29]. We model basis shapes using the Chen-Nguyen (CN11) simulator [26] rather than the GSA [32] for improved accuracy at moderate block sizes.

We adopt the MATZOV cost model [11], used in ML-KEM and ML-DSA parameter selection [4], [6]: for BKZ with block size β, classical bit-security is $\mathsf{Cost}_{\mathsf{MATZOV}}(\beta) = 0.292 \cdot \beta + o(\beta)$, accounting for BDGL sieving [27] with nearest-neighbor optimizations. This improves upon ADPS16 [29] and ABLR21 [33] through refined memory-time tradeoffs.

We evaluate three attacks: primal uSVP [29], which embeds LWE such that $(\mathbf{e}, \mathbf{s}, 1)$ is the unique shortest vector; primal BDD [34], applying Babai's algorithm on a BKZ-reduced basis [35], [36]; and dual hybrid (MATZOV) [30], [11], combining lattice reduction with meet-in-the-middle search and FFT-based distinguishing.

Beyond Gaussian errors, we analyze restricted secret distributions: CBD_η for $\eta \in \{1, 2, 3\}$ (used in ML-KEM), ternary over $\{-1, 0, 1\}$, binary over $\{0, 1\}$, and sparse ternary with Hamming weight h. When advantageous, the Bai-Galbraith embedding [14] incorporates small secrets into the short vector target.

Unless otherwise specified, experiments use: $q = 3329$, discrete Gaussian with $\sigma = 1.22$ (matching CBD_3 variance), $m = d$ samples, MATZOV cost model, and CN11 simulator.

IV. Experimental Analysis

In this section, we go through the experiments that we did using Lattice Estimator [8] (also used in Zama's Concrete Compiler [9]) with the MATZOV cost model, but with some further improvements because some inefficiencies have been reported for Lattice Estimator's CN11 [37]. Our work is conducted on the Apple MacBook Air (2022) with the M2 chip, A15 Bionic (64-bit ARM-based), an 8-core CPU featuring a base clock speed of 3.49 GHz, and 16GB of memory, Tahoe OS 26.2 build 25C56, SageMath v.10.4, and Python v.3.12.3.

A. Rank-Dimension Tradeoff Analysis

Tables II and III present primal uSVP and dual hybrid costs across all 32 configurations.

The main finding in this part is that for each fixed d, *all* (n, k) decompositions yield identical security, β, and lattice

TABLE II: Primal uSVP attack costs. Parameters: $q = 3329$, $\sigma = 1.22$.

d	n	k	Security	β	Lattice dim.
256	256, 128, 64, 32	1, 2, 4, 8	76.3	165	511
512	512, 256, 128, 64, 32	1, 2, 4, 8, 16	143.7	406	985
768	768, 384, 256, 192, 128, 96	1, 2, 3, 4, 6, 8	214.9	660	1451
1024	1024, 512, 256, 128, 64	1, 2, 4, 8, 16	288.5	922	1906
1280	1280, 640, 256, 128	1, 2, 5, 10	363.7	1189	2370
1536	1536, 512, 384, 256	1, 3, 4, 6	440.3	1461	2790

TABLE III: Dual hybrid attack costs (MATZOV). Parameters: $q = 3329$, $\sigma = 1.22$.

d	n	k	Security (bits)	β
256	256, 128, 64, 32	1, 2, 4, 8	75.5	157
512	512, 256, 128, 64, 32	1, 2, 4, 8, 16	139.6	384
768	768, 384, 256, 192, 128, 96	1, 2, 3, 4, 6, 8	206.2	624
1024	1024, 512, 256, 128, 64	1, 2, 4, 8, 16	275.9	872
1280	1280, 640, 256, 128	1, 2, 5, 10	346.3	1120
1536	1536, 512, 384, 256	1, 3, 4, 6	418.9	1381

dimension. At $d = 512$, configurations from $(512, 1)$ to $(32, 16)$ all achieve 143.7 bits (primal) and 139.6 bits (dual). This invariance follows from the unrolling: attacks operate on the d-dimensional LWE instance without exploiting module structure. Practitioners may thus choose (n, k) based on implementation considerations, NTT efficiency, memory, side-channel resistance, without security penalty. While this invariance is consistent with the fact that current lattice attacks on LWE usually do not exploit module structure [38], [39] (except for other problems, e.g., SIS [40]), it had not been verified empirically across a broad range of (n, k) decompositions. Our results confirm that the estimator outputs agree up to its numerical precision, which gives concrete support to the parameter-flexibility assumption underlying the ML-KEM and ML-DSA standards.

B. Security Scaling with Dimension

Figure 1 shows security versus d for primal uSVP, dual hybrid, and primal BDD across $d \in [256, 1536]$.

Security scales linearly: regression yields slope ≈ 0.284 bits/dimension ($R^2 > 0.9999$). The dual hybrid consistently provides the tightest bound, trailing primal uSVP by 4–21 bits. BKZ block sizes range from $\beta = 165$ ($d = 256$) to $\beta = 1461$ ($d = 1536$) for primal attacks. NIST Level 1 requires $d \geq 450$; Level 3 requires $d \geq 680$; Level 5 requires $d \geq 900$. ML-KEM parameters ($d \in \{512, 768, 1024\}$) exceed these thresholds.

C. Impact of Secret Distribution

Table IV quantifies the security loss from restricted secrets relative to a Gaussian baseline.

The main finding here is that CBD_3 incurs < 0.3 bits loss, validating ML-KEM's choice. $\mathrm{CBD}_2, \mathrm{CBD}_1$ costs 3.6–

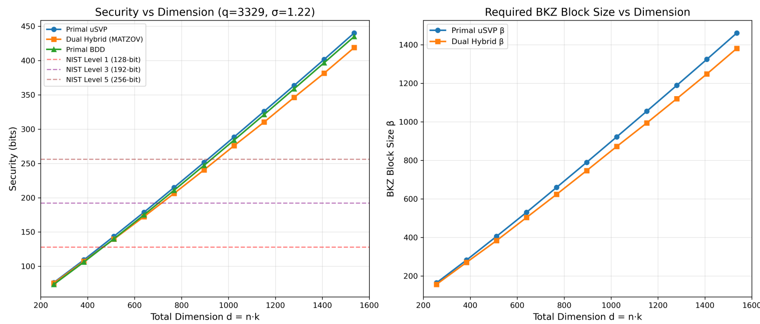

Fig. 1: Security scaling with dimension. Left: Attack costs with NIST level thresholds. Right: Required BKZ block size β.

TABLE IV: Secret distribution impact. Δ denotes the security loss relative to the Gaussian($\sigma = 1.22$) baseline under the primal uSVP attack, which we verified to be the tighter of the two across all rows; hence $\Delta = \min(\Delta_{\text{primal}}, \Delta_{\text{dual}})$. Parameters: $q = 3329$, $\sigma_e = 1.22$.

d	Secret	Primal	Dual	β_{p}	Δ
512	Gaussian(1.22)	143.7	139.6	406	—
512	Gaussian(3.19)	161.9	156.4	472	-18.2
512	CBD_3	143.8	139.7	406	-0.0
512	CBD_2	140.2	135.9	393	$+3.6$
512	CBD_1	134.3	128.4	372	$+9.5$
512	Ternary	136.8	131.5	381	$+7.0$
512	Binary	128.4	120.5	351	$+15.3$
512	Sparse($h = 64$)	128.4	120.5	351	$+15.3$
768	Gaussian(1.22)	214.9	206.2	660	—
768	CBD_3	215.0	206.5	660	-0.0
768	CBD_2	209.6	201.0	641	$+5.3$
768	CBD_1	200.3	189.1	608	$+14.6$
768	Ternary	204.0	193.5	621	$+10.9$
768	Binary	191.1	177.5	575	$+23.8$
768	Sparse($h = 64$)	185.8	171.2	556	$+29.2$
1024	Gaussian(1.22)	288.5	275.9	922	—
1024	CBD_3	288.6	276.0	922	-0.0
1024	CBD_2	281.2	267.4	896	$+7.3$
1024	CBD_1	268.6	252.2	851	$+20.0$
1024	Ternary	273.7	259.6	869	$+14.9$
1024	Binary	255.9	236.0	806	$+32.6$
1024	Sparse($h = 64$)	243.3	223.2	761	$+45.2$
1280	Gaussian(1.22)	363.7	346.3	1189	—
1280	CBD_3	363.9	346.5	1190	-0.3
1280	CBD_2	354.4	335.4	1156	$+9.3$
1280	CBD_1	338.3	317.2	1099	$+25.4$
1280	Ternary	345.1	325.6	1123	$+18.6$
1280	Binary	322.3	295.9	1042	$+41.4$
1280	Sparse($h = 64$)	300.9	277.1	966	$+62.7$

9.3 and 9.5–25.4 bits respectively, scaling superlinearly with d. Binary secrets cost 15.3–41.4 bits due to reduced variance enabling more effective Bai-Galbraith embedding [14], [41]. Sparse ternary ($h = 64$) reaches 62.7-bit penalty at $d = 1280$ as combinatorial attacks exploit decreasing h/d [42]. Larger Gaussian $\sigma = 3.19$ *increases* security by 18–43 bits at the cost of higher decryption failure probability.

D. Overstretched Regime Analysis

Figure 2 shows security degradation as $\log_2(q/\sigma^2)$ increases from 7.4 to 23.4 ($q \in [2^8, 2^{24}]$, $\sigma = 1.22$).

We consider the threshold $\log_2(q/\sigma^2) \approx 15$ directly from Figure 2: it is the point at which the dual hybrid cost drops below the primal uSVP cost and the slope settles to its asymptotic value of 6–8 bits per unit, marking the onset of the regime where dual attacks dominate. Three regimes emerge: secure ($\log_2(q/\sigma^2) \lesssim 10$), transitional (10–15), and overstretched ($\gtrsim 15$). At $d = 512$, security drops from 166.6 bits ($\log_2(q/\sigma^2) = 9.4$) to 71.9 bits (23.4)—a 95-bit reduction at rate 6–8 bits per unit. ML-KEM parameters ($q = 3329$, $\sigma \approx 1.22$, yielding $\log_2(q/\sigma^2) \approx 11.5$) lie safely within the transitional [15], [11] regime.

E. Modulus and Noise Sensitivity

Figure 3 presents parameter sensitivity at $d = 512$. Security decreases logarithmically with q: from 183.6 bits ($q = 512$) to 106.1 bits ($q = 65537$), roughly 10–13 bits per doubling. For noise sensitivity, primal costs increase monotonically (117.5 bits at $\sigma = 0.5$ to 213.0 bits at $\sigma = 5.0$), while dual costs rise sharply for $\sigma \gtrsim 2.5$, becoming infinite at $\sigma = 5.0$. The crossover implies: for small σ (ML-KEM regime), dual hybrid is the binding constraint; for large σ, primal dominates. ML-KEM's $\sigma = 1.22$ achieves near-balanced attack costs.

V. Comparison with NTRU

NTRU [43] security reduces to finding short vectors $f, g \in \mathcal{R}$ in $\Lambda_h^{\text{NTRU}} = \{(\mathbf{u}, \mathbf{v}) \in \mathbb{Z}^{2n} : \mathbf{u} + h \cdot \mathbf{v} \equiv 0 \pmod q\}$, where $h = g \cdot f^{-1} \mod q$. NTRU is vulnerable to sublattice attacks in overstretched regimes [44], [19], while MLWE admits both primal and dual strategies.

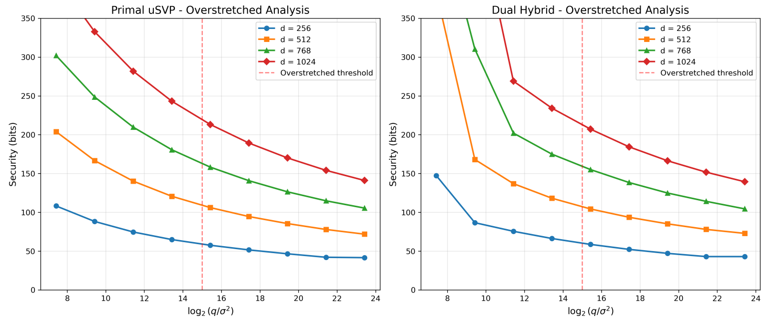

Fig. 2: Overstretched regime: security vs. $\log_2(q/\sigma^2)$. Vertical line at threshold ≈ 15. Parameters: $\sigma = 1.22$.

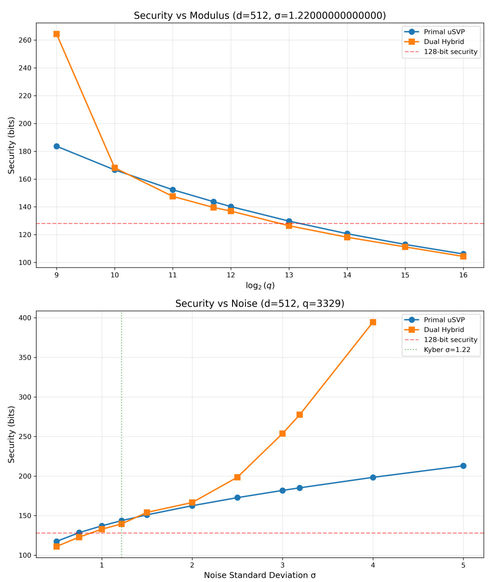

Fig. 3: Top: Security vs. q ($d = 512$, $\sigma = 1.22$). Bottom Security vs. σ ($d = 512$, $q = 3329$). Dotted line is Kyber's $\sigma = 1.22$.

A. Comparative Security Analysis

Table V compares MLWE (Kyber-style: $n = 256$, $k \in \{2, 3, 4\}$, $q = 3329$, CBD) with NTRU (ternary secrets) and Dilithium ($q = 8380417$, CBD). At NIST Level 1, ML-KEM-512 and NTRU-509 achieve comparable security (136.0 vs. 137.3 bits). However, NTRU-677 ($d = 677$) achieves only 181.5 bits, short of the roughly 192-bit target associated with NIST Level 3, while ML-KEM-768 ($d = 768$) achieves 196.4 bits, a 14.9-bit advantage at similar dimensions. NTRU exhibits greater modulus sensitivity: NTRU-821 requires $q = 4096$ and $d = 821$ to reach 204.5-bit security level. Dilithium demonstrates MLWE's flexibility [45], [46] with large moduli ($q = 8380417$) by compensating through increased dimension.

B. Structural Differences

NTRU's dense sublattice attack [19], [20] becomes effective when $q > n^{2.484}$, constraining parameter selection. MLWE distances this vulnerability but faces dual hybrid attacks providing the tightest bounds (4–13 bits below primal). The absence of dual attack estimates for NTRU in Table V reflects that NTRU dual attacks reduce to primal variants without independent improvement.

TABLE V: MLWE vs. NTRU security comparison.

Scheme	d	q	Primal	Dual	β	NIST
ML-KEM-512	512	3329	140.4	136.0	394	L1
ML-KEM-768	768	3329	204.9	196.4	624	L3
ML-KEM-1024	1024	3329	275.1	262.3	874	L5
NTRU-509	509	2048	137.3	—	383	L1
NTRU-677	677	2048	181.5	—	541	L1
NTRU-821	821	4096	204.5	—	622	L3
NTRU-1277	1277	2048	347.1	—	1130	L5
Dilithium-2	1024	8380417	143.5	141.0	400	L1
Dilithium-3	1536	8380417	231.3	226.4	713	L3
Dilithium-5	2048	8380417	302.4	292.6	966	L5

C. Standardization Implications

NIST selected ML-KEM over NTRU-based schemes [47]. Our analysis identifies contributing factors: MLWE offers (n, k) decomposition flexibility without security loss (Section IV-A), enabling fixed $n = 256$ NTTs across security levels; MATZOV [11], [48] provides concrete, validated security bounds; and worst-case reductions [3] provide theoretical assurance absent in NTRU.

VI. Conclusion

We presented a systematic security analysis of MLWE across 32 configurations with total dimensions $d = n \cdot k \in \{256, 512, 768, 1024, 1280, 1536\}$. Our experiments establish that generic lattice attacks depend solely on d, with all (n, k) decompositions yielding identical primal and dual security estimates. For standardized parameters, we confirm: ML-KEM-512 achieves 136.0 bits (dual), ML-KEM-768 achieves 196.4 bits, and ML-KEM-1024 achieves 262.3 bits. The security penalty for CBD_3 secrets is below 0.3 bits, while binary secrets incur a reducion of 15–41 bits. The overstretched threshold occurs at $\log_2(q/\sigma^2) \approx 15$. These results validate the design choices in ML-KEM and provide concrete foundations for future MLWE instantiations.

References

[1] O. Regev, "On lattices, learning with errors, random linear codes, and cryptography," in *Journal of the ACM*, vol. 56, no. 6, 2009, pp. 34:1–34:40, preliminary version in STOC 2005.

[2] V. Lyubashevsky, C. Peikert, and O. Regev, "On ideal lattices and learning with errors over rings," in *EUROCRYPT 2010*, ser. LNCS, vol. 6110. Springer, 2010, pp. 1–23.

[3] A. Langlois and D. Stehlé, "Worst-case to average-case reductions for module lattices," in *Designs, Codes and Cryptography*, vol. 75, no. 3, 2015, pp. 565–599.

[4] National Institute of Standards and Technology, "FIPS 203: Module-lattice-based key-encapsulation mechanism standard," Federal Information Processing Standards Publication, 2024.

[5] J. W. B. et al., "CRYSTALS-Kyber: A CCA-Secure module-lattice-based KEM," in *IEEE EuroS&P*. IEEE, 2018, pp. 353–367.

[6] National Institute of Standards and Technology, "FIPS 204: Module-lattice-based digital signature standard," Federal Information Processing Standards Publication, 2024.

[7] J. W. Bos, C. Costello, L. Ducas, I. Mironov, M. Naehrig, V. Nikolaenko, A. Raghunathan, and D. Stebila, "Frodo: Take off the ring! practical, quantum-secure key exchange from LWE," in *ACM CCS 2016*. ACM, 2016, pp. 1006–1018.

[8] M. R. Albrecht, R. Player, and S. Scott, "On the concrete hardness of Learning with Errors," Journal of Mathematical Cryptology, pp. 169–203, 2015.

[9] Zama, "Concrete: Tfhe compiler that converts python programs into fhe equivalent," 2022, https://github.com/zama-ai/concrete.

[10] A. et al., "Homomorphic encryption security standard," HomomorphicEncryption.org, Toronto, Canada, Tech. Rep., 2018.

[11] MATZOV, "Report on the security of LWE: Improved dual lattice attack," Zenodo, 2022, https://doi.org/10.5281/zenodo.6412487.

[12] L. Ducas and L. N. Pulles, "Does the dual-sieve attack on learning with errors even work?" in *CRYPTO 2023*, ser. LNCS, vol. 14083. Springer, 2023, pp. 37–69.

[13] A. Pouly and Y. Shen, "Provable dual attacks on learning with errors," in *EUROCRYPT 2024*, ser. LNCS, vol. 14657. Springer, 2024, pp. 256–285.

[14] S. Bai and S. D. Galbraith, "Lattice decoding attacks on binary LWE," in *ACISP 2014*, ser. LNCS, vol. 8544. Springer, 2014, pp. 322–337.

[15] M. R. Albrecht, "On dual lattice attacks against small-secret LWE and parameter choices in HElib and SEAL," in *EUROCRYPT 2017*, ser. LNCS, vol. 10211. Springer, 2017, pp. 103–129.

[16] B. Applebaum, D. Cash, C. Peikert, and A. Sahai, "Fast cryptographic primitives and circular-secure encryption based on hard learning problems," in *CRYPTO 2009*, ser. LNCS, vol. 5677. Springer, 2009, pp. 595–618.

[17] H. Chen, I. Chillotti, and Y. Song, "Improved bootstrapping for approximate homomorphic encryption," in *EUROCRYPT 2019*, ser. LNCS, vol. 11477. Springer, 2019, pp. 34–54.

[18] A. Pouly and Y. Shen, "Hardness of LWE on general entropic distributions," *IACR Cryptology ePrint Archive*, vol. 2021, p. 1521, 2021.

[19] P. Kirchner and P.-A. Fouque, "Revisiting lattice attacks on overstretched NTRU parameters," in *EUROCRYPT 2017*, ser. LNCS, vol. 10210. Springer, 2017, pp. 3–26.

[20] L. Ducas and W. P. J. van Woerden, "NTRU fatigue: How stretched is overstretched?" in *ASIACRYPT 2021*, ser. LNCS, vol. 13093. Springer, 2021, pp. 3–32.

[21] M. R. Albrecht and L. Ducas, *Lattice Attacks on NTRU and LWE: A History of Refinements*, ser. London Mathematical Society Lecture Note Series. Cambridge University Press, 2021, pp. 15–40.

[22] M. R. Albrecht, F. Göpfert, F. Virdia, and T. Wunderer, "Revisiting the expected cost of solving uSVP and applications to LWE," in *ASIACRYPT 2017*, ser. LNCS, vol. 10624. Springer, 2017, pp. 297–322.

[23] C. Peikert, "Public-key cryptosystems from the worst-case shortest vector problem," in *STOC*, 2009, pp. 333–342.

[24] M. R. Albrecht, V. Gheorghiu, E. W. Postlethwaite, and J. M. Schanck, "Estimating quantum speedups for lattice sieves," in *ASIACRYPT 2020*, ser. LNCS, vol. 12492. Springer, 2020, pp. 583–613.

[25] C.-P. Schnorr and M. Euchner, "Lattice basis reduction: Improved practical algorithms and solving subset sum problems," in *Mathematical Programming*, vol. 66, 1994, pp. 181–199.

[26] Y. Chen and P. Q. Nguyen, "BKZ 2.0: Better lattice security estimates," in *ASIACRYPT 2011*, ser. LNCS, vol. 7073. Springer, 2011, pp. 1–20.

[27] A. Becker, L. Ducas, N. Gama, and T. Laarhoven, "New directions in nearest neighbor searching with applications to lattice sieving," in *SODA*, 2016, pp. 10–24.

[28] A. et al., "The general sieve kernel and new records in lattice reduction," in *EUROCRYPT 2019*, ser. LNCS, vol. 11477. Springer, 2019, pp. 717–746.

[29] E. Alkim, L. Ducas, T. Pöppelmann, and P. Schwabe, "Post-quantum key exchange – a new hope," in *25th USENIX Security Symposium*, 2016, pp. 327–343.

[30] N. Howgrave-Graham, "A hybrid lattice-reduction and meet-in-the-middle attack against NTRU," in *CRYPTO 2007*, ser. LNCS, vol. 4622. Springer, 2007, pp. 150–169.

[31] Q. Guo and T. Johansson, "Faster dual lattice attacks for solving LWE with applications to CRYSTALS," in *ASIACRYPT 2021*, ser. LNCS, vol. 13093. Springer, 2021, pp. 33–62.

[32] C.-P. Schnorr, "Lattice reduction by random sampling and birthday methods," *ANTS VI*, vol. 3076, pp. 145–156, 2004.

[33] M. R. A. et al., "Estimate all the LWE, NTRU schemes!" in *SCN 2018*, ser. LNCS, vol. 11035. Springer, 2018, pp. 351–367.

[34] R. Lindner and C. Peikert, "Better key sizes (and attacks) for LWE-based encryption," in *CT-RSA 2011*, ser. LNCS, vol. 6558. Springer, 2011, pp. 319–339.

[35] D. Dachman-Soled, L. Ducas, H. Gong, and M. Rossi, "LWE with side information: Attacks and concrete security estimation," in *CRYPTO 2020*, ser. LNCS, vol. 12171. Springer, 2020, pp. 329–358.

[36] E. W. Postlethwaite and F. Virdia, "On the success probability of solving unique SVP via BKZ," in *PKC 2021*, ser. LNCS, vol. 12710. Springer, 2021, pp. 68–98.

[37] "Lattice estimator is very slow for larger lattice dimensions (2^{14} or higher)," GitHub Issue https://github.com/malb/lattice-estimator/issues/75, May 2023, gitHub repository: `malb/lattice-estimator`.

[38] T. Mukherjee and N. Stephens-Davidowitz, "Lattice reduction for modules, or how to reduce ModuleSVP to ModuleSVP," in *CRYPTO 2020*, ser. LNCS, vol. 12171. Springer, 2020, pp. 213–242.

[39] C. Lee, A. Pellet-Mary, D. Stehlé, and A. Wallet, "An LLL algorithm for module lattices," in *ASIACRYPT 2019*, ser. LNCS, vol. 11922. Springer, 2019, pp. 59–90.

[40] L. Ducas, T. Espitau, and E. W. Postlethwaite, *Finding Short Integer Solutions When the Modulus Is Small*. Springer Nature Switzerland, 2023, p. 150–176. [Online]. Available: http://dx.doi.org/10.1007/978-3-031-38548-3_6

[41] L. Chua, H. Chen, Y. Song, and K. Lauter, "On the concrete security of LWE with small secret," 2024, springer, open access.

[42] C. Li, J. Sotáková, E. Wenger, M. Malhou, E. Garcelon, F. Charton, and K. E. Lauter, "SALSA PICANTE: A machine learning attack on LWE with binary secrets," in *ACM CCS 2023*. ACM, 2023, pp. 2606–2620.

[43] J. Hoffstein, J. Pipher, and J. H. Silverman, "NTRU: A ring-based public key cryptosystem," in *ANTS III*, ser. LNCS, vol. 1423. Springer, 1998, pp. 267–288.

[44] M. R. Albrecht, S. Bai, and L. Ducas, "A subfield lattice attack on overstretched NTRU assumptions," in *CRYPTO 2016*, ser. LNCS, vol. 9814. Springer, 2016, pp. 153–178.

[45] L. D. et al., "CRYSTALS-Dilithium: A lattice-based digital signature scheme," *IACR Transactions on Cryptographic Hardware and Embedded Systems*, vol. 2018, no. 1, pp. 238–268, 2018.

[46] D. et al., "Saber: Module-LWR based key exchange, CPA-secure encryption and CCA-secure KEM," in *AFRICACRYPT 2018*, ser. LNCS, vol. 10831. Springer, 2018, pp. 282–305.

[47] F. et al., "Falcon: Fast-Fourier lattice-based compact signatures over NTRU," in *Submission to NIST Post-Quantum Cryptography Standardization*, 2020, https://falcon-sign.info/falcon.pdf.

[48] K. Carrier, C. Meyer-Hilfiger, Y. Shen, and J.-P. Tillich, "Assessing the impact of a variant of MATZOV's dual attack on Kyber," in *CRYPTO 2025*, ser. LNCS, vol. 16000. Springer, 2025.